LLM 显存占用分析

训练与推理过程中的内存需求与优化

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Why Memory Matters

A simple problem

假如我有一张 80GB 的显卡，我想训练/推理一个 4B 的模型，我应该设置多大的 batch size 和 sequence length?

在这个 tutorial 中，我们将基于这个问题来进行思考和分析。我们考虑更一般的问题形式：

Motivation

在训练和推理时 LLM 所需要的内存是多少？如何进行优化内存占用？

我们将先介绍训练/推理阶段的内存计算，再针对可优化部分进行分析并介绍相应优化算法。

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Transformer Architecture

以 Qwen3[YLY+25] 为例，现代 LLM 的架构如右图所示。其中具体的模块不同的模型可能有改动。

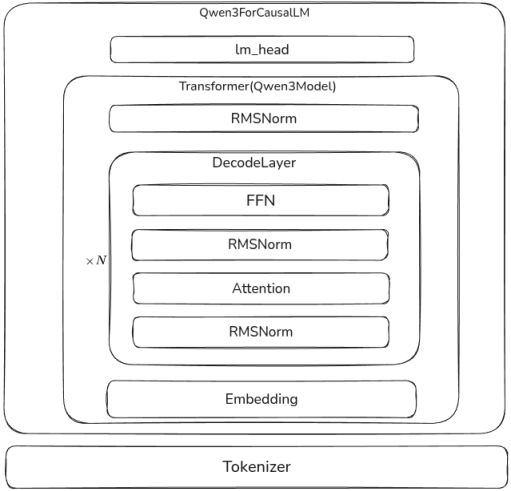

图: Architecture of Qwen3

Notation

与参数量、FLOPs 计算所用记号一致；参数量 P 的推导见 LLM parameter analysis。

变量	含义
P	number of parameters
L	layers
V	vocabulary size
d	hidden size
d_{ff}	FFN hidden size
s	sequence length
b	batch size
h	number of attention heads
d_h	attention head dimension

Assumption

1. 若无特别说明，使用 **BF16/FP16**，每个参数 2 byte。
2. 不使用 dropout（与现代大模型设定一致）。
3. Attention 基于原始 multi-head attention.
4. FFN 基于 SwiGLU.

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Training Memory Componenets

训练部分的内存占用由四部分组成，如下所示

$$
\begin{aligned}
\text{training_memory} &= \text{weight} && (1)\\
&+ \text{activation} && (2)\\
&+ \text{optimizer} && (3)\\
&+ \text{gradient} && (4)
\end{aligned}
$$

Weights: $\boxed{2P}$.

Gradients（与权重同精度）: $\boxed{2P}$.

Optimizer states

AdamW 优化器：

- **一阶动量**：$2P$,
- **二阶动量**：$2P$,
- **合计**：$2 \times 2P = \boxed{4P}$.

Algorithm 1: AdamW[LH19]

Input: $\alpha,\ \lambda,\ \beta_1, \beta_2 \in [0, 1),\ \epsilon,\ \theta_0$

Output: θ_t

$m_0 \leftarrow 0,\ v_0 \leftarrow 0,\ t \leftarrow 0;$

while *Not converge* **do**

$t \leftarrow t + 1;$

$g_t \leftarrow \nabla_\theta f_t(\theta_{t-1});$

$m_t \leftarrow \beta_1 \cdot m_{t-1} + (1 - \beta_1) \cdot g_t;$

$v_t \leftarrow \beta_2 \cdot v_{t-1} + (1 - \beta_2) \cdot g_t^2;$

$\hat{m}_t \leftarrow \frac{m_t}{1-\beta_1^t};$

$\hat{v}_t \leftarrow \frac{v_t}{1-\beta_2^t};$

$\theta_t \leftarrow \theta_{t-1} - \alpha \left(\frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} + \lambda \theta_{t-1} \right);$

Activation

Motivation

激活值是前向传播过程中计算得到的中间结果，用于在反向传播时计算梯度。

我们仅针对 linear layer 进行推导

forward
$$\mathbf{z}_\ell = W_\ell \mathbf{a}_{\ell-1} + b_\ell, \quad \mathbf{a}_\ell = \phi(\mathbf{z}_\ell)$$

backward
$$\frac{\partial \mathcal{L}}{\partial W_\ell} = \frac{\partial \mathcal{L}}{\partial \mathbf{z}_\ell} \cdot \frac{\partial \mathbf{z}_\ell}{\partial W_\ell} = \frac{\partial \mathcal{L}}{\partial \mathbf{z}_\ell} \boxed{\mathbf{a}_{\ell-1}}$$

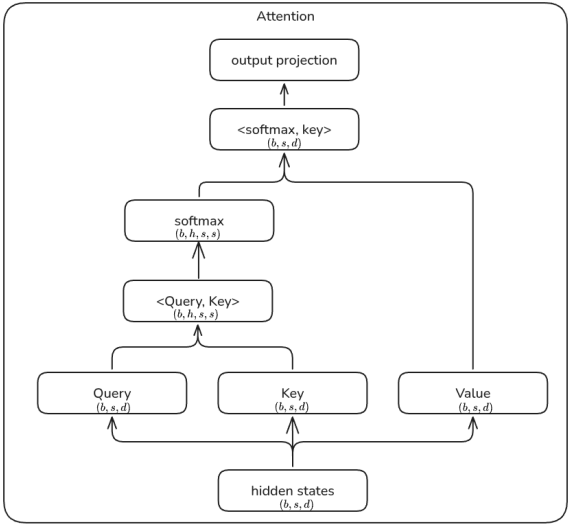
可以看到，计算第 ℓ 层关于 W_ℓ 的梯度时需要其输入 $\mathbf{a}_{\ell-1}$，因此训练时需保存每个模块对应的输入，也就是激活值 (activation).

Activation-Attention

按计算图（无优化）可得需保存的激活：

- Q/K/V 投影：共享输入 $\to 2bsd$
- $Q^\top K$：Q, K 均需保存 $\to 2 \times 2bsd = 4bsd$
- softmax 输入：$2bhs^2$
- weighted sum 输入：$2bhs^2 + 2bsd$
- output projection 输入：$2bsd$

Attention 合计：$\boxed{10sbd + 4bhs^2}$.

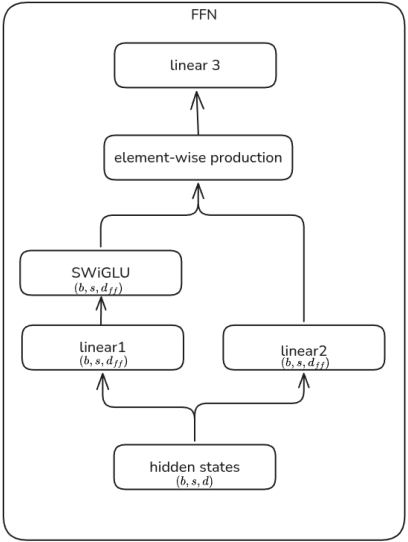


Activation-FFN & LayerNorm

FFN（SwiGLU，assume $d_{\mathrm{ff}} = 4\,d$）:

- 第一层输入 $2\,sbd$
- SwiGLU 输入 $2 \times d_{ff} \times d = 8\,sbd$
- 第二层输入 $2 \times d_{ff} \times d = 8\,sbd$
- 合计：$\boxed{18sbd}$

LayerNorm：保存输入 $\rightarrow$ $\boxed{2bsd}$.

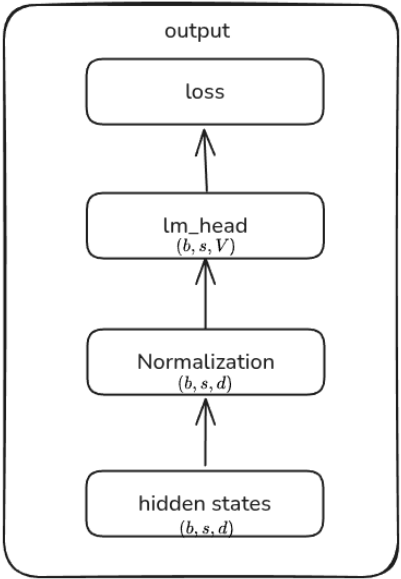


Activation-Output

Output 包含以下组成部分：

- **FinalNorm** 输入：$2bsd$
- **lm_head** 输入：$2bsd$
- **Loss** 输入：$2bsV$

合计：$\boxed{4bsd + 2bsV}$。



Activation-Total

我们将上面的结果汇总在一起，得到[1]

$$
\begin{aligned}
\text{activation} &= L \cdot \text{transformer_block} + \text{output} && (5)\\
&= L \cdot (\text{Pre_Norm} + \textit{Attention} + \text{Post_Norm} + \text{FFN}) + \text{output} && (6)\\
&= \boxed{bs(32dL + 4hsL + 4d + 2V)} && (7)\\
&\approx bsL(32d + 4hs) && (8)\\
&\approx 4bs^2hL && (9)
\end{aligned}
$$

我们可以看到，未优化的情况下，$\text{activation} \propto bs^2$. 这里 s^2 主要由 attention 部分产生，后续 flash attenion 就针对这一点进行了优化。

[1] Qwen3 中，$2V/32dL = 6\%$, $32dL/4hsL = 2.5\%$.

Total Training Memory

我们将上面的结果进行汇总得到

$$
\begin{aligned}
\text{training_memory} &= \text{weight} + \text{activation} + \text{optimizer} + \text{gradient} && (10)\\
&= 2P + bs(32dL + 4hsL + 4d + 2V) + 4P + 2P && (11)\\
&= 8P + bs(32dL + 4hsL + 4d + 2V) && (\text{exact})\\
&\approx 8P + 4bs^2hL && (12)
\end{aligned}
$$

可以看到，训练阶段的内存占用分为固定部分 ($8P$) 和动态部分 ($4bs^2hL$)，动态部分主要是 attention 的缓存。

Analysis

我们以 Qwen3 模型为例，得到如下热力图，batchh size 越小，说明 activation 占用越高

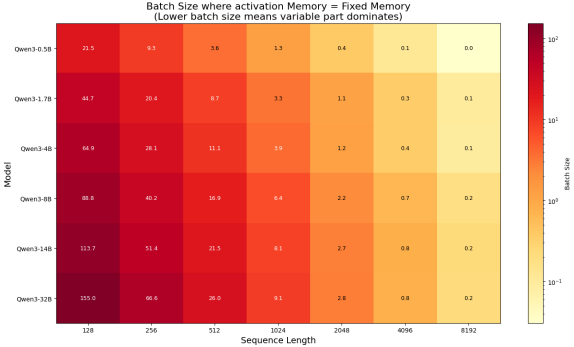


图: Comparison of activation memory with fixed memory

Experiments

我们分别针对 80GB 的显卡计算 Qwen3 系列模型的最高配置：

Model	P	L	h	s	predicted b	actual b
Qwen3-0.6B	0.6	28	16	512	68	34
Qwen3-1.7B	1.7	28	16	512	42	28
Qwen3-4B	4	36	32	512	16	12
Qwen3-8B	8.1	36	32	512	4	2

其中 predicted b 基于前面的准确公式计算得到；actual b 通过实验验证得到。（注意我们这里的 prediction 没有考虑任何优化手段与其他内存开销，因此与实际值有出入）

Case Study

我们分别使用 Qwen3-4B 和 Qwen3-8B 来进行实验 ($b = 1$, $s = 512$), 结果如下[2]

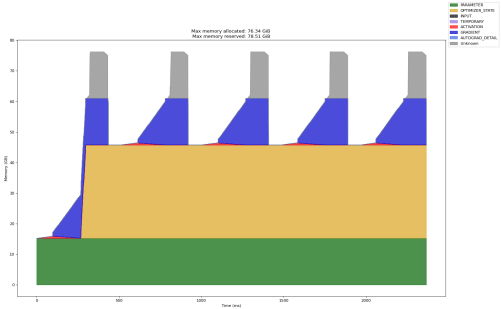


图: Memory profile of Qwen3-8B

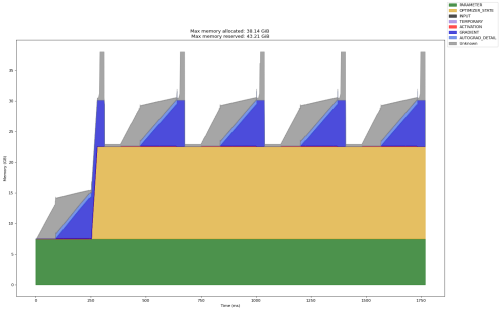


图: Memory profile of Qwen3-4B

[2]参考PyTorch 显存可视化与 Snapshot 数据分析

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Inference Components

Inference 阶段内存占用主要与模型参数，KV cache 两部分相关。

$$\text{Inference_Memory} = \text{weight} \tag{13}$$
$$+ \text{activation} \tag{14}$$
$$+ \text{KV cache} \tag{15}$$

- Weights：$\boxed{2P}$
- Activations（经验[3]，batch size=1）：$\approx 0.4P$
- KV cache：与序列长度相关，见后续分析。

[3] 见 https://blog.eleuther.ai/transformer-math/

KV Cache Mechanism

Motivation

LLM 推理中为避免重复计算历史 token 的 key/value 而使用的**空间换时间**的缓存机制。

自回归时逐 token 生成，每步 attention 形式为（$\mathbf{q}_t$ 当前 query，$\mathbf{k}_{:,t}/\mathbf{v}_{:,t}$ 历史 K/V）：

$$\mathbf{q}_t = W_Q\mathbf{x}_t, \quad \mathbf{k}_{:,t} = W_K[\mathbf{x}_1, \ldots, \mathbf{x}_t], \quad \mathbf{v}_{:,t} = W_V[\mathbf{x}_1, \ldots, \mathbf{x}_t]$$

处理下一 token $\mathbf{x}_{t+1}$ 时只需在已有结果后追加当前步：

$$\mathbf{k}_{:,t+1} = W_K[\mathbf{x}_1, \ldots, \mathbf{x}_{t+1}] = [\mathbf{k}_{:,t},\ W_K\mathbf{x}_{t+1}], \quad \mathbf{v}_{:,t+1} = W_V[\mathbf{x}_1, \ldots, \mathbf{x}_{t+1}] = [\mathbf{v}_{:,t},\ W_V\mathbf{x}_{t+1}]$$

缓存前，每生成一个 token 都重新计算 $\rightarrow$ 总计算量 $\sum_{t=1}^{s} \mathcal{O}(t) = \mathcal{O}(s^2)$；

缓存后，每步只算当前 token $W_K\mathbf{x}_{t+1}$、$W_V\mathbf{x}_{t+1}$ $\rightarrow$ 计算量 $\mathcal{O}(s)$、空间占用 $\mathcal{O}(s)$ ——**以空间换时间**。

KV Cache Memory

对于 multi-head attention ，KV cache 的显存占用为：

$$\text{Memory(KV cache)} = s \times 2 \times 2 \times L \times h \times d_h = \boxed{4sLhd_h} \tag{16}$$

因子含义： s 序列长，2 为 K+V, L 层、h 头、d_h 头维度。

Remark

- KV 占用与**模型配置**（L, h, d_h）和**序列长度** s 都有关，token 越多占用越高。
- 实际中因 *page granularity*、*padding*、*fragmentation* 往往略高于理论值。
- 长输出时 KV 占比会超过权重，成为推理瓶颈 → 见后续「KV Cache 优化」。

Inference Memory

综合前面分析，我们可以得到推理阶段的总内存为：

$$\boxed{\text{Inference_Memory} \approx 2.4P + 4sLhd_h}$$

可以看到，推理阶段也由固定部分（参数量，activation）以及动态部分（KV cache）组成。

Experiments

我们仍然使用 Qwen3 系列来进行实验（未考虑任何优化，仅考虑 KV cache 和 prefilling），结果如下：

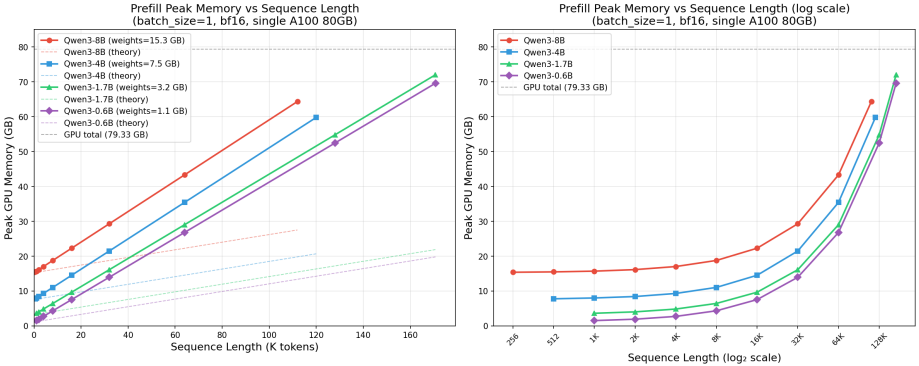

图: Memory profile for Qwen3 series

注意这里的 peak memory 实际上并不能和 KV cache 的显存占用完全对应，因为 peak memory 还包括了 attention 的临时显存等。

Dynamic v.s. Static

接下来我们看一下模型权重与 KV cache 的显存占用情况。由于 Qwen3 的 KV cache 计算为 $4sLh_{kv}d_h$，而不同模型只有 L 不一样，因此对于更大的模型，KV cache 显存占用超过模型权重的上下文长度更高。

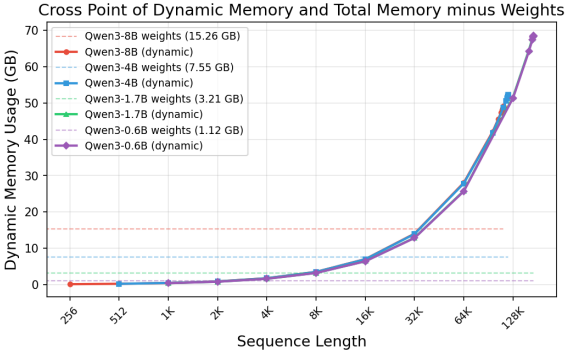

图: Dynamic vs. Static memory profile for Qwen3 series

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Overview

阶段	核心方法	典型技术
Training	显存与效率提升	Activation Checkpointing Mixed Precision Training Flash Attention ZeRO Pipeline/Model/Data Parallelism
Inference	长序列与速度优化	KV Cache Optimization Paged/Radix Attention Faster Attention Quantization

Mixed Precision Training

Motivation

计算量大的部分用低精度，计算量小的部分用高精度。低精度参与运算，高精度避免 Overflow/Underflow.

下图是 DeepSeek-V3[DALF+25] 使用的混合精度训练框架

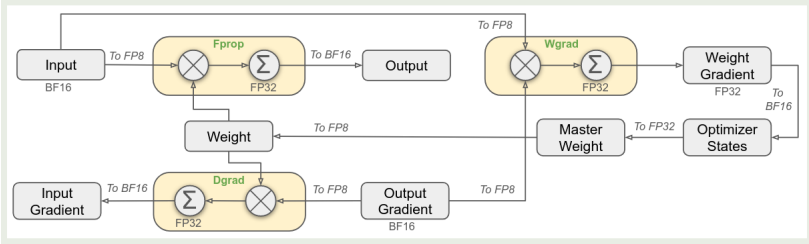


图: Mixed precision training of DeepSeek-V3

Mixed Precision Training-Analysis

Precision	**BF16**	**FP32**	**BF16**
AMP	No	Yes	Yes
Weights	BF16 (2)	FP32 (4)	BF16 (2)
Master weigths	-	-	FP32 (4)
Gradients	BF16 (2)	FP32 (4)	BF16 (2)
Adam m	BF16 (2)	FP32 (4)	FP32 (4)
Adam v	BF16 (2)	FP32 (4)	FP32 (4)
Static total (bytes/param)	8	16	16

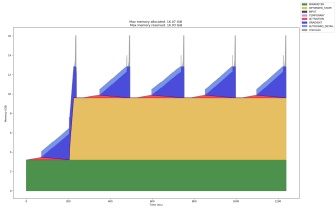
表: Comparison of memory usage per parameter across different training configurations.

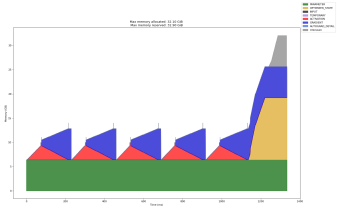
Remark

1. BF16 (w/ AMP) 与 FP32 (w/ AMP) 的静态显存占用相同，但 BF16 (w/ AMP) 的动态显存占用更低。
2. 主流框架基本都使用了 BF16/FP8 (w/ AMP) 的训练方式。

Mixed Precision Training-Experiments

Experiments on Qwen3-1.7B, batch size $= 1$, sequence length $= 512$.





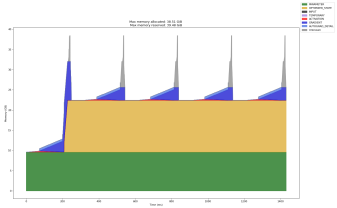


图: (left) BF16 (w/o AMP), $\sim 8P$ bytes, (center) FP32 (w/ AMP), $\sim 16P$ bytes, (right) BF16 (w/ AMP), $\sim 16P$ bytes

ZeRO[RRRH20a]

Motivation

将 optimizer states / gradients / weights 按不同 GPU 切片存储，需要参与计算时再 all-gather 整合成完整参数。这样每张卡只需维护自己负责的一部分，大幅降低单卡显存需求。

ZeRO 的工作原理如下图所示

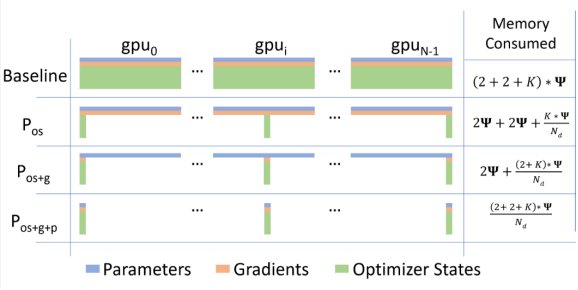


图: ZeRO DP framework

ZeRO Stages

- **ZeRO1**: shard optimizer states

$$\text{Training_Memory} = \text{weight} + \text{activation} + \frac{\text{optimizer}}{\#\text{GPUs}} + \text{gradient} \tag{ZeRO1}$$

- **ZeRO2**: shard gradients + gradients

$$\text{Training_Memory} = \text{weight} + \text{activation} + \frac{\text{optimizer} + \text{gradient}}{\#\text{GPUs}} \tag{ZeRO2}$$

- **ZeRO3**: shard all

$$\text{Training_Memory} = \text{activation} + \frac{\text{weight} + \text{optimizer} + \text{gradient}}{\#\text{GPUs}} \tag{ZeRO3}$$

Remark

ZeRO3 可极大降低单卡显存上限，但通信量也会提高。

Model Parallelism[SPP+20]

Motivation

将模型切分到不同的 GPU 上，计算时，先 dispatch, 再执行计算，最后通过 all-gather 等操作得到最终结果。切分方式包括 PP (Pipeline Parallelism), TP (Tensor Parallelism), EP (Expert Parallelism) 等。

$$\text{Training_Memory} = \frac{\text{Memory(weight)}}{\text{PP degree} \times \text{TP degree}} \tag{17}$$

结合 ZeRO1 与 Model Parallelism 时（activation 中与 TP 相关的部分按 TP degree 缩减）:

$$\text{Memory}_{\text{train}} \approx \frac{\text{weight}}{\text{PP} \times \text{TP}} + \frac{\text{activation}}{\text{TP}} + \frac{\text{optimizer}}{\#\text{GPUs}} + \frac{\text{gradient}}{\text{PP}}$$

Activation Checkpointing[KCL+22]

Motivation

在反向传播时，重新计算所需的输入，来达到以时间换空间的目的。

	No ckpt	Selective ckpt	Full ckpt
memory	很高	中等	很低 $\sim 2bsd$
extra compute	无	中等	很高 $\sim 2Pbs$

Remark

实践中常结合 model parallelism 与 selective checkpointing 来实现 trade off.

Flash Attention[DFE+22]

Motivation

通过将 Attention 的计算进行分块，来提高内存访问效率以及降低反向传播时所需要的 activation 大小。

Flash Attention 通过 tiling 与 online-softmax 降低该部分显存并提升效率[4]。这样 attention 部分的显存就由 $\text{activation} \propto bs^2$ 降低到了 $\text{activation} \propto bs$.

Theorem

flash attention 输出 $O = \mathrm{softmax}(QK^T)V$ (correctness).
其时间复杂度为 $\mathcal{O}(s^2 d)$, 空间复杂度为 $\mathcal{O}(s)$ (memory savings).

[4]详见 notes on Flash Attention

KV Cache Optimization[LLT+25]

$$\text{Memory(KV cache)} = s \times 2 \times 2 \times L \times h \times d_h$$

1. s: KV cache compression, eviction, selection
2. 2: KV cache quantization
3. 2: key-value sharing, MLA[DALF+24]
4. $h \times d_h$: attention, MQA[Sha19], GQA[ALTdJ+23], MLA.

Weight Quantization

Motivation

使用低精度来表示高精度数值的方法，来减少内存占用/提高计算效率。

量化时机	代表性工作
训练后量化 (PTQ)	GPTQ[FAHA23] AWQ[LTT+24] SmoothQuant[XLS+24] GGUF[Ger23]
量化感知训练 (QAT)	LLM-QAT[LOZ+23] PEQA[KLK+23]

表: 大模型量化方法总结

Activation Offloading

Motivation

将一部分参数/优化器状态/激活值等存储到 CPU 上，需要的时候再加载到 GPU 上。

Offloading 场景	代表性工作
训练阶段 Offloading	ZeRO-Offload[RRA+21] / ZeRO-Infinity FSDP[ZGV+23] CPU Offload
推理阶段 Offloading	FlexGen[SZY+23] vLLM[KLZ+23] KV Cache Offload
MoE Offloading	KTransformers[CXZ+25] DeepSpeed-MoE[RLY+22]

表: offloading methods

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Training

Framework	Memory Optimizations
Megatron-LM[SPP+20]	TP, SP, Activation Checkpointing
DeepSpeed[RRRH20b]	ZeRO-1/2/3, CPU/NVMe Offload, Activation Checkpointing
FSDP[ZGV+23]	Full parameter sharding, Gradient sharding, CPU Offload
Colossal-AI[LLB+23]	ZeRO, TP, PP, Activation Checkpointing

Inference

Framework	Key Techniques	Memory Optimizations
vLLM[KLZ+23]	Paged Attention	KV cache paging, Continuous batching
SGLang[ZYX+24]	Radix Attention	KV cache reuse, Efficient scheduling
TensorRT-LLM[Cor23]	Kernel fusion	Weight quantization, KV cache optimization

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Takeaway

Components	Training	Inference	Optimization
weights	$2P$	$2P$	quantization
optimizer states	$4P$	0	ZeRO, offloading
gradients	$2P$	0	ZeRO
activations	$\sim 4Lhs^2b$	$\sim 0.4P$	ckpt, offloading, flash attention
KV cache	0	$4sLhd_h$	kv optimization, attention
TOTAL	$8P + 4Lhs^2b$	$2.4P + 4sLhd_h$	

- **训练**：瓶颈主要在激活值（随 batch size / 序列长度线性增长）。
- **推理**：瓶颈主要在 KV Cache（随序列长度增长）。

Conclusion & Future Directions

Conclusion

We discussed how to analyze the memory occupied by an LLM and how to optimize it.

Future Directions

1. More efficient architecture (attention, MoE).
2. Scalable training/inference framework
3. Software-hardware co-design algorithms.

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